



CS602-FALL
ASSIGNMENT-1
JUNAID MALIK



Junaidfazal08@gmail.com
Bc190202640@vu.edu.pk

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VULMSHELP.COM

JUNAID MALIK
0304-1659294



CS619 & CS519 PROJECTS

پروجیکٹ اور اسٹریٹج
ساتھ ساتھ

Daily Classes & Viva Preparation

Our Expertise

- ✓ Python (Django & Flask)
- ✓ Php (Laraval)
- ✓ MySql

Projects Phase

- ✓ SRS + DD Prototype
- ✓ Final Deliverable



LMS Handling

+92 304 1659294

ajtechinstitute24@gmail.com

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QUESTION NO 1:

```
#include <graphics.h>
```

```
#include <conio.h>
```

```
#include <string>
```

```
#include <sstream>
```

```
void displayText(int x, int y, const std::string& text) {  
    char buffer[100];  
    strncpy(buffer, text.c_str(), sizeof(buffer));  
    buffer[sizeof(buffer) - 1] = '\0'; // Ensure null termination  
    outtextxy(x, y, buffer);  
}
```

```
int main() {  
    // Initialize the graphics mode  
    int gd = DETECT, gm;  
    initgraph(&gd, &gm, NULL); // Use NULL for the default path  
  
    // Student ID (e.g., "ID: JUNAID MALIK")  
    std::string studentID = "ID: JUNAID MALIK";
```

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```
// Daily calorie intake variables
```

```
int Monday = 2200;
```

```
int Tuesday = 1800;
```

```
int Wednesday = 2500;
```

```
int Thursday = 2000;
```

```
int Friday = 2300;
```

```
// Array for days and calorie values
```

```
const char* days[] = {"Monday", "Tuesday", "Wednesday", "Thursday",  
"Friday"};
```

```
int calories[] = {Monday, Tuesday, Wednesday, Thursday, Friday};
```

```
// Calculate total and average calories
```

```
int totalCalories = Monday + Tuesday + Wednesday + Thursday + Friday;
```

```
float averageCalories = totalCalories / 5.0;
```

```
// Display student ID in the top right corner
```

```
displayText(getmaxx() - 200, 10, studentID);
```

```
// Set bar colors for each day
```

```
int colors[] = {RED, GREEN, BLUE, YELLOW, CYAN};
```

```
// Define bar chart properties
```

```
int barWidth = 50;
```

```
int gap = 60;
```

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```
int maxCalories = 3000; // Setting a max limit for height scaling

int x = 100; // Starting x-coordinate for the first bar

int yBase = 400; // Base y-coordinate (ground level of bars)


// Draw the bars
for (int i = 0; i < 5; i++) {
    int barHeight = (calories[i] * 300) / maxCalories; // Scale bar height
    setfillstyle(SOLID_FILL, colors[i]);
    bar(x, yBase - barHeight, x + barWidth, yBase); // Draw the bar


    // Display calorie values above each bar
    std::stringstream ss;
    ss << calories[i];
    displayText(x + 10, yBase - barHeight - 20, ss.str());


    // Display day labels below each bar
    displayText(x + 5, yBase + 10, days[i]);


    // Move x-coordinate for the next bar
    x += barWidth + gap;
}


// Display total and average calories
std::stringstream totalText, avgText;
totalText << "Total Calories: " << totalCalories;
```

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```
avgText << "Average Calories: " << averageCalories;
```

```
displayText(100, 450, totalText.str());
```

```
displayText(100, 470, avgText.str());
```

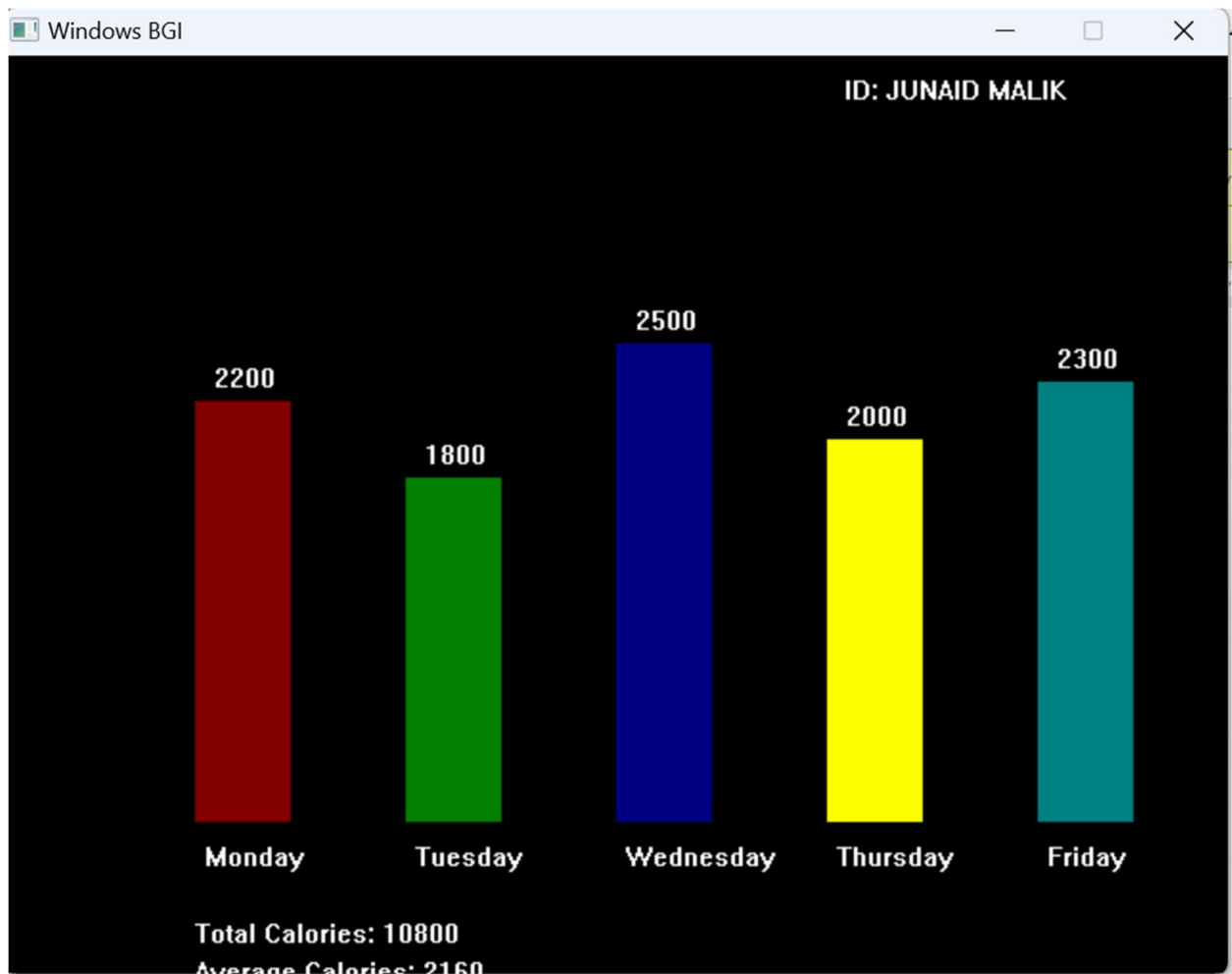
```
getch(); // Wait for user input to close
```

```
closegraph(); // Close graphics mode
```

```
return 0;
```

```
}
```

OUT PUT:



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